

BT505 Paper

mcqs

"A sensor that integrates a biological element with a physiochemical transducer to produce an electronic signal proportional to a single analyte which is then conveyed to a detector."

Ion-selective electrodes Also known as indicator electrodes

-Gas Sensors: Thin film of electrolyte solution is placed between electrode and membrane

-CO₂ Sensors Electrolyte between electrode and membrane is NaHCO₃-NaCl solution

Oxygen Sensors Consists of a pair of electrodes (Ag anode and Pt cathode) in an electrolyte solution

SOLID-STATE DEVICES

is Known as ion-selective field effect transistors (ISFET) Liquid membrane is a type of ISE based on water-immiscible liquid substances produced in a polymeric membrane used for direct potentiometric measurement.

ENFET (Enzyme Field Effect Transistor)

Enzyme thermistors are biosensors that use thermal resistors to measure the heat change caused by an enzymatic reaction

Immunosensors: Use antibody as biorecognition element.

Assay format strongly influenced by:

Nature of analyte

Transduction approach

Secondary antibody (reporter); usually Pam against the primary antibody.

A BioSignal is a human body variable that can be measured and monitored and that can provide information on the health status of individual. In most cases it is an electric voltage

Electric signals

ENG – electroneurogram EEG – electroencephalogram ERG – electroretinogram

ECG – electrocardiogram EMG – electromyogram

EOG – electrooculogram GSR – galvanic skin response

Magnetic biosignals

MNG – magnetoneurogram MEG – magnetoencephalogram MCG –

magnetocardiogram MMG – magnetomyogram

Acoustic biosignals

Subset of mechanical signals that describe the acoustic sound produced by the body

Potentiometric biosensor also used for immunoassays or geno_analysis.

95% of protein nitrogen is metabolized to urea nitrogen

RIfS biosensor for the detection within a concentration range of 1000–100000

cells/ml-1

Evanescent wave aptamer-based Fluorescence biosensor Used for the rapid:
Sensitive Highly selective detection of 17β -oestradiol,
Endocrine-disrupting compound frequently detected in environmental water
samples.

This biosensor was constructed as a portable system with the detection
Bioluminescent optical fibre Response of cells to genotoxic agents

Inorganic anisotropic materials are following quartz crystal ,

A SERS-active surface applied to the sensitive detection of cancer proteins
(~100 pg) In a low sample volume (~10 nl).

Questions:

Pressure Sensor 10

Piezoelectric biosensor

Reverse Piezoelectric effect

Surface enhanced raman scattering biosensors

Evanescent wave fluorescence

Factors needs for MEMS

how are chemical and mechanical biosignals produced

Redox Enzyme

○ **VU BIOLOGISTS**